

Squared or Cubed? — Measurement Unit Detective Answer Key

Grade 5

Quick Answers

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|---|--|
| 1. 28 cm^2 | 6. 24000 cm^3 |
| 2. 24 cm^3 | 7. (b) corrected answer = 20000 cm^2 , — |
| 3. — | 8. (a) small cube = 8 cm^3 , (b) large cube = 64 cm^3 , (c) how many times bigger = 8, — |
| 4. 30 cm^3 | |
| 5. (b) corrected answer = 36 cm^3 , — | |
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What is verified

4 of 8 problems fully machine-checked against SymPy.
— marks an answer only you can judge — problems 3, 5, 7, 8. The worked solution states what a correct response must contain.

Curriculum

Level: Grade 5

4.MD – problems 1, 3

5.MD – problems 2, 3, 4, 5, 6, 7, 8

Difficulty 1–5 of 5 across 15 tagged checks.

Common wrong answers

1. *If they got 22: found the perimeter (added the sides) instead of the area*
 2. *If they got 6: counted the cubes in one layer only*
 4. *If they got 15: multiplied only two dimensions (found the area of the base)*
 6. *If they got 90: added the three dimensions instead of multiplying them*
 7. *If they got 200: used the length conversion factor only once*
 8. *If they got 6: multiplied the edge by 3 instead of using it as a factor three times*
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1. Sticker: rectangle 7 cm by 4 cm — area

Solution: Area covers a flat surface with unit *squares*: $7 \times 4 = 28$ of them fit. The unit is square centimeters.

28 cm^2

2. Box packed with centimeter cubes: 6 cubes per layer, 4 layers

Solution: Volume counts unit *cubes*: 6 cubes in each layer, 4 layers, so $6 \times 4 = 24$ cubes fill the box.

24 cm ³

3. Pick the best unit (cm, m², mL, or cm³)

Solution: (a) A classroom floor is a flat surface, so it takes square units: *square meters* (m²).

(b) Juice fills space inside the glass, so it takes a capacity or cubic unit: *milliliters* (mL) — cubic centimeters (cm³) also correct.

(c) A pencil's length runs along one dimension: *centimeters* (cm).

4. Cereal box $5 \times 3 \times 2$ cm — volume

Solution: $V = \ell \times w \times h = 5 \times 3 \times 2$. The base holds $5 \times 3 = 15$ cubes and there are 2 layers, giving 30 cubes.

30 cm ³

5. Tomas's jewelry box: " $6 \times 2 \times 3 = 36$ cm²"

Solution: (a) The number 36 is correct, but the *unit* is wrong: volume counts unit cubes filling the box, so it needs *cubic* centimeters. Square centimeters measure a flat surface. A correct response must say the unit should be cubed, not squared.

(b) Corrected: $6 \times 2 \times 3 = 36$, in cubic units.

36 cm ³

6. Aquarium $40 \times 20 \times 30$ cm

Solution: $V = 40 \times 20 \times 30$. Step by step: $40 \times 20 = 800$ for the base, times 30 layers gives 24,000.

24000 cm ³

7. Lena's conversion: " 2 m² = $2 \times 100 = 200$ cm²"

Solution: (a) Lena used the *length* conversion once, but square units need it *twice*: a square meter is 100 cm by 100 cm. A correct response must say the factor of 100 has to be applied twice (or that 1 m² = 10000 cm²).

(b) Corrected: $2 \times 100 \times 100 = 20,000$.

20000 cm ²

8. Challenge: cubes with edges 2 cm and 4 cm

Solution: (a) $V = 2 \times 2 \times 2 = 8$.

8 cm ³

(b) $V = 4 \times 4 \times 4 = 64$.

64 cm ³

(c) $64 \div 8 = 8$, so the large volume is times the small one.

8

(d) Roy doubled the volume once, but the edge is a factor *three* times: length, width, and height all doubled, so the volume is multiplied by $2 \times 2 \times 2 = 8$, not by 2. A correct

response must say all three dimensions doubled.